INTERNATIONAL CYBERSECURITY AND DIGITAL FORENSICS ACADEMY

LABORATORY INVESTIGATION & FORENSIC EVIDENCE REPORT

"Preparing Global Cybersecurity Leaders Through Excellence in Hands-On Training"

Lab Title: SBT-DF203 Lab 1: HTTP Analysis Using Wireshark – Text Traffic	**Course Name:** Basic Networking Skills for Digital Forensics
Student Name: Almustapha Yusuf	**Registration Number:** fwsd2511343
Operating Platform: Kali Linux 2026 (Rolling Release)	**Date of Submission:** 07/09/2026
Forensic Scope: Localhost Loopback Capture (lo)	**Report Version:** Version 1.0 (Final Release)

Executive Summary

This report documents the forensic analysis of a controlled plaintext HTTP session. A local Apache webpage was created containing the student's identity. TShark was used to capture loopback traffic on TCP port 80. The capture was systematically analyzed to identify the TCP three-way handshake, the plaintext HTTP GET request, the HTTP 200 OK response, and the four-step connection closure sequence. The report details the encapsulation of evidentiary data across the Ethernet, IP, TCP, and HTTP layers and provides an in-depth forensic analysis regarding the limitations of loopback captures concerning physical Layer 2 MAC addresses. Dual-copy SHA-256 cryptographic hashes were calculated to guarantee chain-of-custody integrity.

1. Authority, Scope, and Evidence Description

- **Authority:** This investigation was conducted within an isolated, strictly authorized Kali Linux 2026 virtual laboratory. No external, unapproved, or third-party networks or systems were targeted.
- **Forensic Scope:** Controlled forensic capture and packet dissection of a single plaintext HTTP GET session directed at http://127.0.0.1/basic.html.
- **Primary Evidence File:** evidence/basic.pcapng (Original bitstream network capture preserved under write-protected permissions).
- **Working Forensic Copy:** working/basic_working.pcapng (Bit-identical duplicate utilized for active packet dissection and filtering).
- **Cryptographic Hash Digest (SHA-256):** 4fb3910cb6127763a6367512e74688e3ed779c345b4f0da1e83f51d4a79d6e3 (Computed across both files, proving zero data alteration).

2. Environment and Acquisition Method

- **Host Environment:** Kali Linux 2026 (Rolling Release, 64-bit).
- **Web Server Daemon:** Apache/2.4.68 (Debian), active and listening on TCP port 80 (PID: 804085).
- **Capture Engine:** TShark / Wireshark 3.2.1 sniffing on the Loopback (lo) interface with display and capture filters.
- **HTTP Client / Traffic Generator:** cURL 8.21.0 executed with explicit flags (--no-keepalive -v) to enforce synchronous teardown.

- **Capture Execution Workflow:** Traffic captured using: sudo tshark -i lo -f 'tcp port 80' -w evidence/basic.pcapng. The capture stopped cleanly following the completed 10-packet HTTP transaction.

3. Chronological Packet Timeline

The 10 captured frames represent a complete, self-contained HTTP conversational transaction on loopback:

#	Timestamp	Source	Destination	Flags	Seq	Ack	Description
1	16:33:36.981510	127.0.0.1:34664	127.0.0.1:80	SYN	0	0	Client initiates connection
2	16:33:36.981757	127.0.0.1:80	127.0.0.1:34664	SYN, ACK	0	1	Server confirms & opens reverse
3	16:33:36.981770	127.0.0.1:34664	127.0.0.1:80	ACK	1	1	Client completes 3-way handshake
4	16:33:36.982120	127.0.0.1:34664	127.0.0.1:80	PSH, ACK	1	1	HTTP GET /basic.html (83 B)
6	16:33:36.985286	127.0.0.1:80	127.0.0.1:34664	PSH, ACK	1	84	HTTP/1.1 200 OK (381 B)
8	16:33:36.986258	127.0.0.1:34664	127.0.0.1:80	FIN, ACK	84	382	Client initiates connection close
9	16:33:36.986491	127.0.0.1:80	127.0.0.1:34664	FIN, ACK	382	85	Server acknowledges & closes

4. Handshake Analysis

The Transmission Control Protocol (TCP) three-way handshake establishes a reliable, bi-directional virtual circuit before any HTTP application payload is transmitted:

- **Packet 1 (SYN):** The client initiates connection from ephemeral port 34664 to Apache port 80. Flag: SYN (0x0002), relative Seq: 0, Ack: 0.
- **Packet 2 (SYN-ACK):** The Apache web server acknowledges the connection request and provides its own synchronization sequence. Flags: SYN, ACK (0x0012), relative Seq: 0, Ack: 1.
- **Packet 3 (ACK):** The client responds to the server's sequence, completing the three-way handshake. Flags: ACK (0x0010), relative Seq: 1, Ack: 1. The socket state transitions to ESTABLISHED.

5. HTTP Request/Response Reconstruction

Reconstruction of the full TCP session stream via tshark -q -z follow,tcp,ascii,0 reveals the raw application conversation without encryption or obfuscation:

```
GET /basic.html HTTP/1.1
Host: 127.0.0.1
User-Agent: curl/8.21.0
Accept: */*

HTTP/1.1 200 OK
```

```
Date: Mon, 07 Sep 2026 20:33:36 GMT
Server: Apache/2.4.68 (Debian)
Last-Modified: Mon, 07 Sep 2026 20:19:35 GMT
ETag: "82-65aea5637199b"
Accept-Ranges: bytes
Content-Length: 130
Vary: Accept-Encoding
Content-Type: text/html

<!DOCTYPE html>
<html><body><h1>SBT-DF203 HTTP Evidence</h1>
<p>Name: Almustapha Yusuf</p>
<p>Reg No: fwsd2511343</p></body></html>
```

6. Encapsulation and MAC Address Explanation

Protocol dissection of Packet 4 (HTTP GET Request) and Packet 6 (HTTP 200 OK Response) provides empirical evidence of multi-layer OSI encapsulation:

Layer / Metric	Packet 4 (GET Request)	Packet 6 (200 OK Response)
Frame Length (Layer 2)	149 bytes	447 bytes
IP Total Length (Layer 3)	135 bytes	433 bytes
TCP Header Length (Layer 4)	32 bytes (8 words)	32 bytes (8 words)
TCP Application Payload (Layer 7)	83 bytes	381 bytes (Header + 130 B HTML)

- **Loopback MAC Address Anomaly (00:00:00:00:00:00):** In loopback captures, both source and destination MAC addresses are populated with all zeroes (00:00:00:00:00:00). Because loopback packets travel purely through kernel memory without ever reaching a physical Network Interface Card (NIC), Address Resolution Protocol (ARP) is bypassed. As a result, synthetic pseudo-headers are populated by the operating system kernel.

7. Forensic Evidence Screenshots

The following 15 documented figures depict the complete, granular lab workflow from environment configuration to deep packet inspection of the request and response trees:

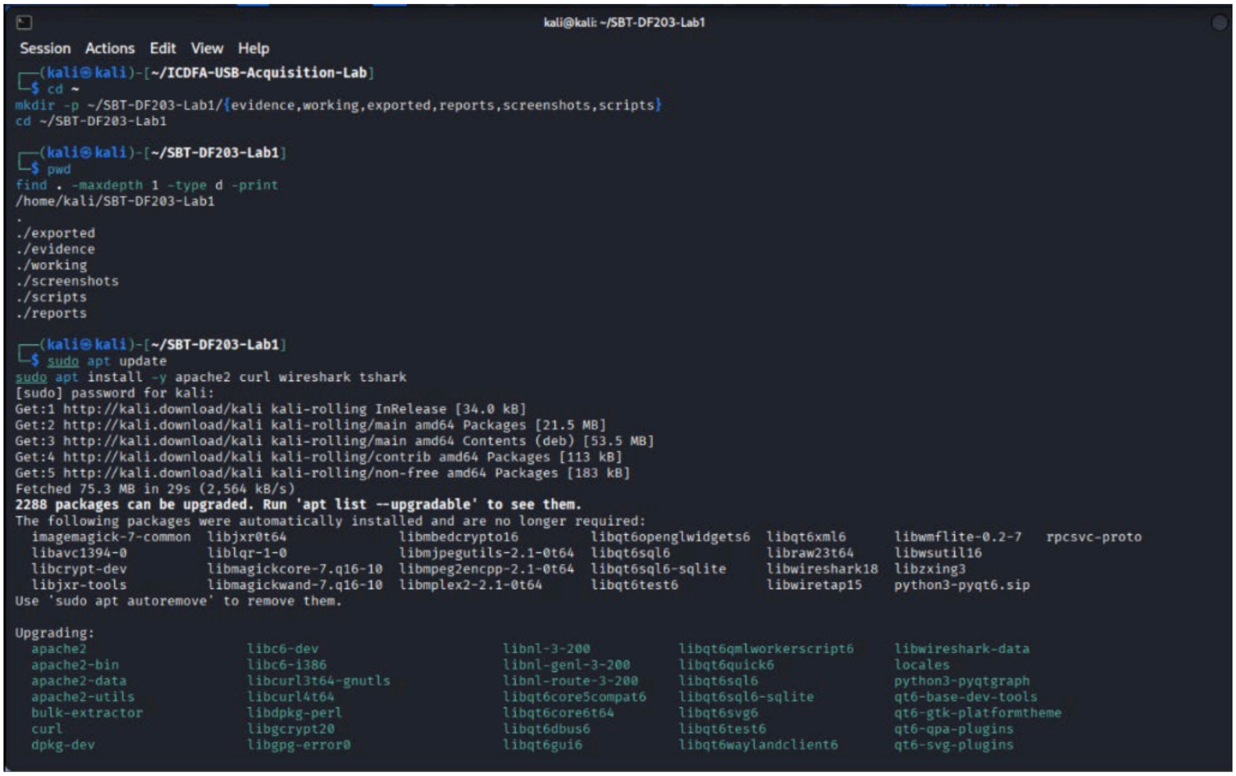


Figure 1: *Workspace Initialization & Dependency Installation* — Creating directory hierarchy and updating APT repositories to install Apache2, cURL, and TShark.

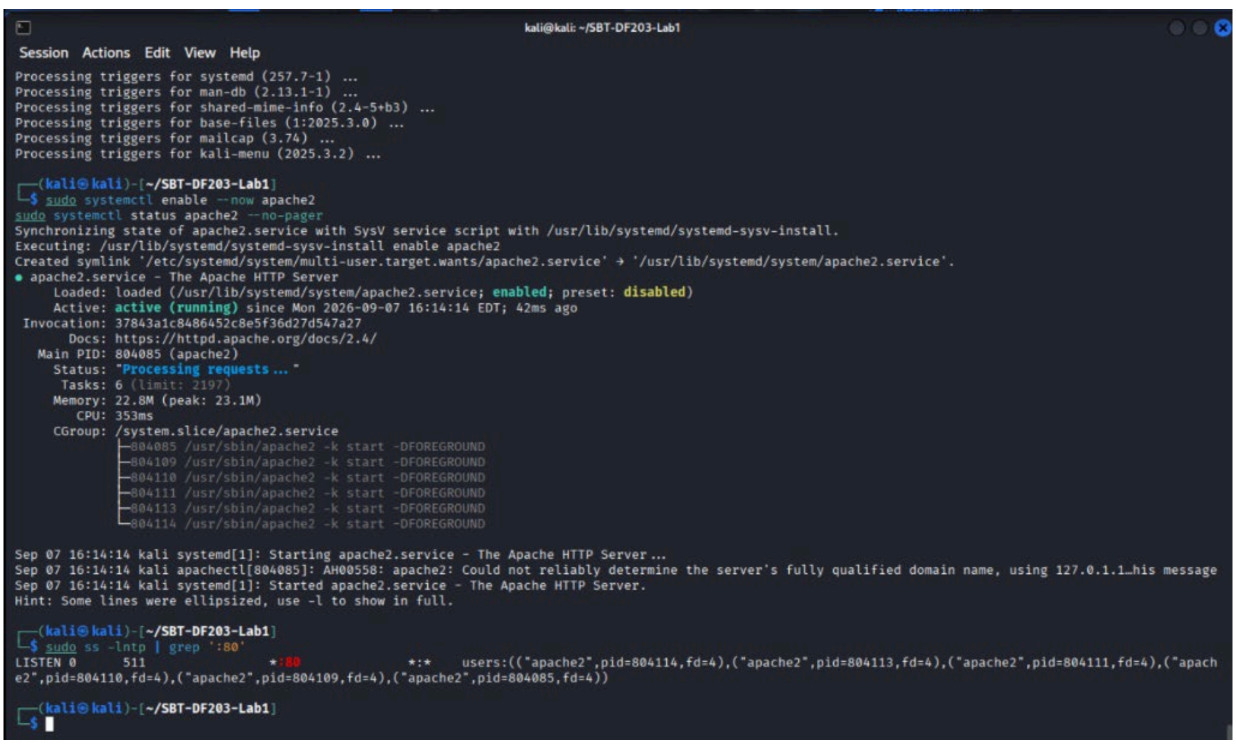


Figure 2: *Apache Web Server Deployment & Listening State Confirmation* — Enabling apache2.service and verifying listening state on TCP port 80 via ss -lntp.

```
┌──(kali㉿kali)-[~/SBT-DF203-Lab1]
└─$ cd ~/SBT-DF203-Lab1
printf '<!DOCTYPE html>\n<html><body><h1>SBT-DF203 HTTP Evidence</h1><p>Name: Almustapha Yusuf</p><p>Reg No: fwsd2511343</p></body></html>\n' | sudo tee /va
r/www/html/basic.html
<!DOCTYPE html>
<html><body><h1>SBT-DF203 HTTP Evidence</h1><p>Name: Almustapha Yusuf</p><p>Reg No: fwsd2511343</p></body></html>

┌──(kali㉿kali)-[~/SBT-DF203-Lab1]
└─$ cat /var/www/html/basic.html
<!DOCTYPE html>
<html><body><h1>SBT-DF203 HTTP Evidence</h1><p>Name: Almustapha Yusuf</p><p>Reg No: fwsd2511343</p></body></html>

┌──(kali㉿kali)-[~/SBT-DF203-Lab1]
└─$ curl -v http://127.0.0.1/basic.html 2>&1 | tee reports/curl_verbose.txt
*   Trying 127.0.0.1:80 ...
* Established connection to 127.0.0.1 (127.0.0.1 port 80) from 127.0.0.1 port 34138
  % Total    % Received % Xferd  Average Speed  Time    Time    Time   Current
                                 Dload  Upload  Total   Spent   Left   Speed
  0      0   0      0   0      0      0      0                              0* using HTTP/1.x
> GET /basic.html HTTP/1.1
> Host: 127.0.0.1
> User-Agent: curl/8.21.0
> Accept: */*
>
* Request completely sent off
< HTTP/1.1 200 OK
< Date: Mon, 07 Sep 2026 20:20:22 GMT
< Server: Apache/2.4.68 (Debian)
< Last-Modified: Mon, 07 Sep 2026 20:19:35 GMT
< ETag: "82-65aea5637199b"
< Accept-Ranges: bytes
< Content-Length: 130
< Vary: Accept-Encoding
< Content-Type: text/html
<
{ [130 bytes data]
100    130 100    130   0      0  18294      0                              0
* Connection #0 to host 127.0.0.1:80 left intact
<!DOCTYPE html>
<html><body><h1>SBT-DF203 HTTP Evidence</h1><p>Name: Almustapha Yusuf</p><p>Reg No: fwsd2511343</p></body></html>

┌──(kali㉿kali)-[~/SBT-DF203-Lab1]
```

Figure 3: *HTML Evidence Page Authoring & Pre-Capture Curl Verification* — Generating /var/www/html/basic.html with student credentials and testing curl response.

```
┌──(kali㉿kali)-[~/SBT-DF203-Lab1]
└─$ cat reports/curl_verbose.txt
*   Trying 127.0.0.1:80 ...
* Established connection to 127.0.0.1 (127.0.0.1 port 80) from 127.0.0.1 port 34138
  % Total    % Received % Xferd  Average Speed  Time    Time    Time   Current
                                 Dload  Upload  Total   Spent   Left   Speed
  0      0   0      0   0      0      0      0                              0* using HTTP/1.x
> GET /basic.html HTTP/1.1
> Host: 127.0.0.1
> User-Agent: curl/8.21.0
> Accept: */*
>
* Request completely sent off
< HTTP/1.1 200 OK
< Date: Mon, 07 Sep 2026 20:20:22 GMT
< Server: Apache/2.4.68 (Debian)
< Last-Modified: Mon, 07 Sep 2026 20:19:35 GMT
< ETag: "82-65aea5637199b"
< Accept-Ranges: bytes
< Content-Length: 130
< Vary: Accept-Encoding
< Content-Type: text/html
<
{ [130 bytes data]
100    130 100    130   0      0  18294      0                              0
* Connection #0 to host 127.0.0.1:80 left intact
<!DOCTYPE html>
<html><body><h1>SBT-DF203 HTTP Evidence</h1><p>Name: Almustapha Yusuf</p><p>Reg No: fwsd2511343</p></body></html>

┌──(kali㉿kali)-[~/SBT-DF203-Lab1]
└─$ █
```

Figure 4: *Verbose HTTP Response Inspection* — Reviewing HTTP headers and raw 130-byte payload output inside curl_verbose.txt.

```
┌──(kali㉿kali)-[~/SBT-DF203-Lab1]
└─$ cd ~/SBT-DF203-Lab1

┌──(kali㉿kali)-[~/SBT-DF203-Lab1]
└─$ sudo tshark -i lo -f 'tcp port 80' -w /home/kali/SBT-DF203-Lab1/evidence/basic.pcapng
Running as user "root" and group "root". This could be dangerous.
Warning: program compiled against libxml 215 using older 214
Capturing on 'Loopback: lo'
tshark: The file to which the capture would be saved ("/home/kali/SBT-DF203-Lab1/evidence/basic.pcapng") could not be opened: Permission denied.

┌──(kali㉿kali)-[~/SBT-DF203-Lab1]
└─$ sudo usermod -aG wireshark kali
# Log out and log back in (or run 'newgrp wireshark')
tshark -i lo -f 'tcp port 80' -w /home/kali/SBT-DF203-Lab1/evidence/basic.pcapng
Warning: program compiled against libxml 215 using older 214
Capturing on 'Loopback: lo'
10 ^C

┌──(kali㉿kali)-[~/SBT-DF203-Lab1]
└─$ cd ~/SBT-DF203-Lab1
cp --preserve=timestamp evidence/basic.pcapng working/basic_working.pcapng
sha256sum evidence/basic.pcapng working/basic_working.pcapng | tee reports/capture_hashes.txt
4fb3910cb712677c3a6367512e74688e3ed779c345b4f0da1e83f51d4a79d6e3  evidence/basic.pcapng
4fb3910cb712677c3a6367512e74688e3ed779c345b4f0da1e83f51d4a79d6e3  working/basic_working.pcapng

┌──(kali㉿kali)-[~/SBT-DF203-Lab1]
└─$ 
```

Figure 5: *TShark Capture Execution & SHA-256 Hashing* — Capturing 10 loopback packets, duplicating to working copy, and computing matching SHA-256 digests.

```
Session  Actions  Edit  View  Help
kali@kali: ~/SBT-DF203-Lab1    kali@kali: ~/SBT-DF203-Lab1
┌──(kali㉿kali)-[~/SBT-DF203-Lab1]
└─$ cd ~/SBT-DF203-Lab1
curl --no-keepalive -v http://127.0.0.1/basic.html 2>&1 | tee reports/curl_capture.txt
*   Trying 127.0.0.1:80...
* Established connection to 127.0.0.1 (127.0.0.1 port 80) from 127.0.0.1 port 34664
  % Total    % Received % Xferd  Average Speed  Time    Time     Time  Current
                                 Dload  Upload  Total   Spent    Left  Speed
  0     0    0     0    0     0      0      0                              0* using HTTP/1.x
> GET /basic.html HTTP/1.1
> Host: 127.0.0.1
> User-Agent: curl/8.21.0
> Accept: */*
>
* Request completely sent off
< HTTP/1.1 200 OK
< Date: Mon, 07 Sep 2026 20:33:36 GMT
< Server: Apache/2.4.68 (Debian)
< Last-Modified: Mon, 07 Sep 2026 20:19:35 GMT
< ETag: "82-65aea5637199b"
< Accept-Ranges: bytes
< Content-Length: 130
< Vary: Accept-Encoding
< Content-Type: text/html
<
{ [130 bytes data]
100    130 100    130    0     0  25415      0                              0
* Connection #0 to host 127.0.0.1:80 left intact
<!DOCTYPE html>
<html><body><h1>SBT-DF203 HTTP Evidence</h1><p>Name: Almustapha Yusuf</p><p>Reg No: fwsd2511343</p></body></html>

┌──(kali㉿kali)-[~/SBT-DF203-Lab1]
└─$ 
```

Figure 6: *Controlled HTTP Client Generation via cURL* — Issuing curl request with --no-keepalive flag to enforce immediate TCP session teardown.

```
┌──(kali㉿kali)-[~/SBT-DF203-Lab1]
└─$ tshark -r working/basic_working.pcapng -Y 'tcp.stream eq 0 && tcp.flags.syn == 1' -T fields -e frame.number -e frame.time -e ip.src -e tcp.srcport -e ip
.dst -e tcp.dstport -e tcp.flags -e tcp.seq -e tcp.ack | tee reports/handshake_packets.tsv
Warning: program compiled against libxml 215 using older 214
1       2026-09-07T16:33:36.981510340-0400      127.0.0.1       34664   127.0.0.1       80      0x0002  0       0
2       2026-09-07T16:33:36.981757334-0400      127.0.0.1       80      127.0.0.1       34664   0x0012  0       1

┌──(kali㉿kali)-[~/SBT-DF203-Lab1]
└─$ tshark -r working/basic_working.pcapng -Y 'tcp.stream eq 0' -T fields -e frame.number -e frame.time -e ip.src -e tcp.srcport -e ip.dst -e tcp.dstport -e
 tcp.flags -e tcp.seq -e tcp.ack | tee reports/full_tcp_stream.tsv
Warning: program compiled against libxml 215 using older 214
1       2026-09-07T16:33:36.981510340-0400      127.0.0.1       34664   127.0.0.1       80      0x0002  0       0
2       2026-09-07T16:33:36.981757334-0400      127.0.0.1       80      127.0.0.1       34664   0x0012  0       1
3       2026-09-07T16:33:36.981770008-0400      127.0.0.1       34664   127.0.0.1       80      0x0010  1       1
4       2026-09-07T16:33:36.982120901-0400      127.0.0.1       34664   127.0.0.1       80      0x0018  1       1
5       2026-09-07T16:33:36.982262902-0400      127.0.0.1       80      127.0.0.1       34664   0x0010  1       84
6       2026-09-07T16:33:36.985286944-0400      127.0.0.1       80      127.0.0.1       34664   0x0018  1       84
7       2026-09-07T16:33:36.985363814-0400      127.0.0.1       34664   127.0.0.1       80      0x0010  84      382
8       2026-09-07T16:33:36.986258188-0400      127.0.0.1       34664   127.0.0.1       80      0x0011  84      382
9       2026-09-07T16:33:36.986491448-0400      127.0.0.1       80      127.0.0.1       34664   0x0011  382     85
10      2026-09-07T16:33:36.986670060-0400      127.0.0.1       34664   127.0.0.1       80      0x0010  85      383

┌──(kali㉿kali)-[~/SBT-DF203-Lab1]
└─$ tshark -r working/basic_working.pcapng -Y 'http.request' -T fields -e frame.number -e frame.time -e ip.src -e tcp.srcport -e ip.dst -e tcp.dstport -e ht
tp.request.method -e http.request.uri -e http.host -e http.user_agent | tee reports/http_request.tsv
Warning: program compiled against libxml 215 using older 214
4       2026-09-07T16:33:36.982120901-0400      127.0.0.1       34664   127.0.0.1       80      GET     /basic.html     127.0.0.1       curl/8.21.0

┌──(kali㉿kali)-[~/SBT-DF203-Lab1]
└─$ tshark -r working/basic_working.pcapng -Y 'http.response' -T fields -e frame.number -e frame.time -e ip.src -e tcp.srcport -e ip.dst -e tcp.dstport -e h
ttp.response.code -e http.content_type | tee reports/http_response.tsv
Warning: program compiled against libxml 215 using older 214
6       2026-09-07T16:33:36.985286944-0400      127.0.0.1       80      127.0.0.1       34664   200     text/html

┌──(kali㉿kali)-[~/SBT-DF203-Lab1]
└─$ tshark -r working/basic_working.pcapng -Y 'tcp.flags.fin == 1 || tcp.flags.reset == 1' -T fields -e frame.number -e frame.time -e ip.src -e tcp.srcport
-e ip.dst -e tcp.dstport -e tcp.flags -e tcp.seq -e tcp.ack | tee reports/connection_close.tsv
Warning: program compiled against libxml 215 using older 214
8       2026-09-07T16:33:36.986258188-0400      127.0.0.1       34664   127.0.0.1       80      0x0011  84      382
9       2026-09-07T16:33:36.986491448-0400      127.0.0.1       80      127.0.0.1       34664   0x0011  382     85
```

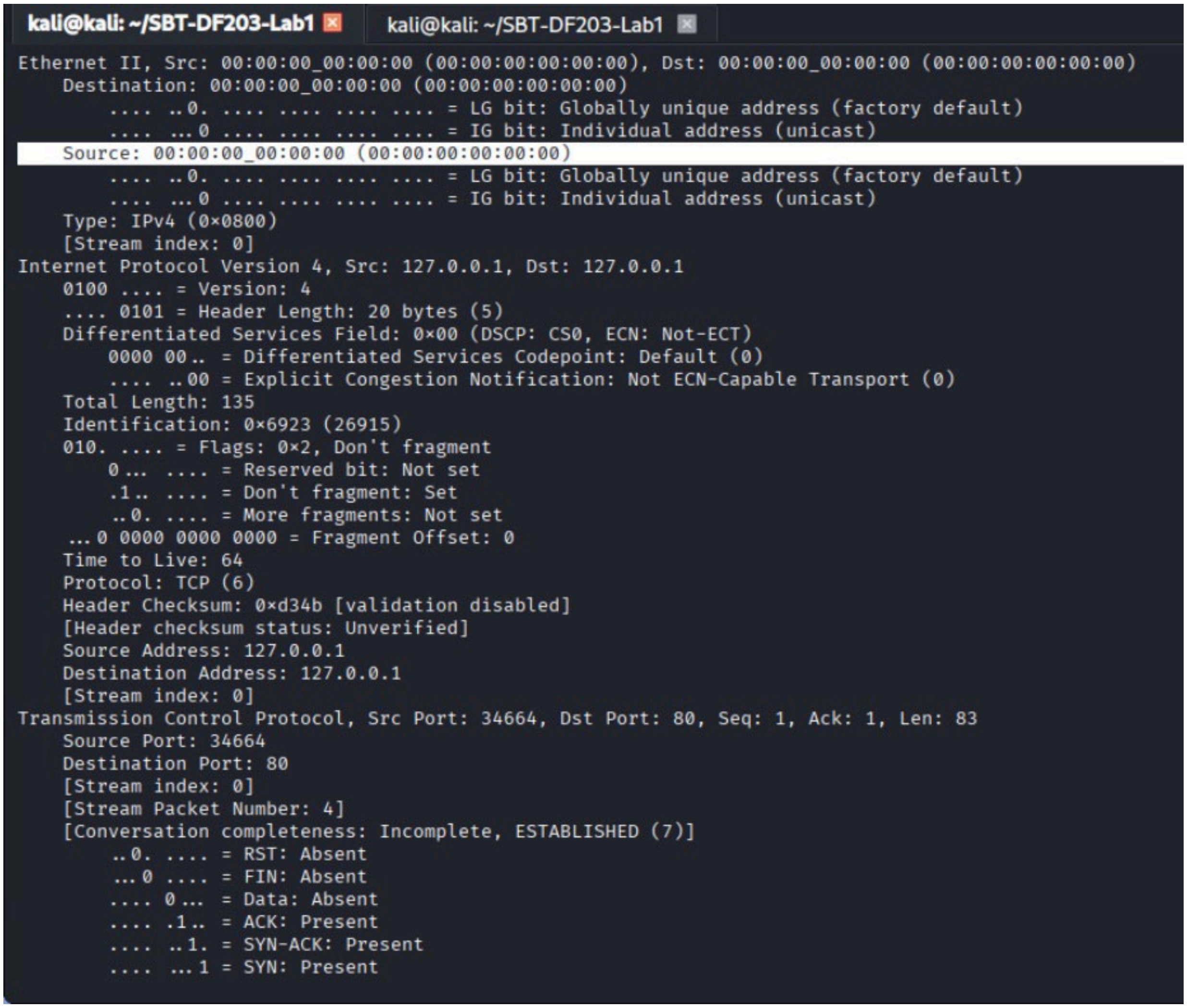
Figure 7: *TShark Chronological Stream Dissection* — Filtering SYN, HTTP request/response, and FIN handshake teardown packets via TShark fields.

```
┌──(kali㉿kali)-[~/SBT-DF203-Lab1]
└─$ tshark -r working/basic_working.pcapng -q -z follow,tcp,ascii,0 | tee reports/follow_tcp_stream.txt
Warning: program compiled against libxml 215 using older 214

===================================================================
Follow: tcp,ascii
Filter: tcp.stream eq 0
Node 0: 127.0.0.1:34664
Node 1: 127.0.0.1:80
83
GET /basic.html HTTP/1.1
Host: 127.0.0.1
User-Agent: curl/8.21.0
Accept: */*


        381
HTTP/1.1 200 OK
Date: Mon, 07 Sep 2026 20:33:36 GMT
Server: Apache/2.4.68 (Debian)
Last-Modified: Mon, 07 Sep 2026 20:19:35 GMT
ETag: "82-65aea5637199b"
Accept-Ranges: bytes
Content-Length: 130
Vary: Accept-Encoding
Content-Type: text/html

<!DOCTYPE html>
<html><body><h1>SBT-DF203 HTTP Evidence</h1><p>Name: Almustapha Yusuf</p><p>Reg No: fwsd2511343</p></body></html>

===================================================================

┌──(kali㉿kali)-[~/SBT-DF203-Lab1]
└─$ tshark -r working/basic_working.pcapng -Y 'http.request || http.response' -T fields -e frame.number -e frame.len -e ip.len -e tcp.len -e tcp.hdr_len -e
http.content_length | tee reports/encapsulation.tsv
Warning: program compiled against libxml 215 using older 214
4       149     135     83      32
6       447     433     381     32      130

┌──(kali㉿kali)-[~/SBT-DF203-Lab1]
└─$ tshark -r working/basic_working.pcapng -Y 'tcp.stream eq 0' -T fields -e frame.number -e eth.src -e eth.dst | tee reports/mac_addresses.tsv
```

Figure 8: *TCP Conversation Follow & Layer Header Length Calculations* — Reconstructing plaintext conversation and computing Frame, IP, and TCP header byte sizes.

```
┌──(kali㉿kali)-[~/SBT-DF203-Lab1]
└─$ tshark -r working/basic_working.pcapng -Y 'tcp.stream eq 0' -T fields -e frame.number -e eth.src -e eth.dst | tee reports/mac_addresses.tsv
Warning: program compiled against libxml 215 using older 214
1       00:00:00:00:00:00       00:00:00:00:00:00
2       00:00:00:00:00:00       00:00:00:00:00:00
3       00:00:00:00:00:00       00:00:00:00:00:00
4       00:00:00:00:00:00       00:00:00:00:00:00
5       00:00:00:00:00:00       00:00:00:00:00:00
6       00:00:00:00:00:00       00:00:00:00:00:00
7       00:00:00:00:00:00       00:00:00:00:00:00
8       00:00:00:00:00:00       00:00:00:00:00:00
9       00:00:00:00:00:00       00:00:00:00:00:00
10      00:00:00:00:00:00       00:00:00:00:00:00

┌──(kali㉿kali)-[~/SBT-DF203-Lab1]
└─$ tshark -r working/basic_working.pcapng -Y 'http.request || http.response' -V | tee reports/http_headers_full.txt
Warning: program compiled against libxml 215 using older 214
Frame 4: Packet, 149 bytes on wire (1192 bits), 149 bytes captured (1192 bits) on interface lo, id 0
    Section number: 1
    Interface id: 0 (lo)
        Interface name: lo
        Interface description: Loopback
    Encapsulation type: Ethernet (1)
    Arrival Time: Sep  7, 2026 16:33:36.982120901 EDT
    UTC Arrival Time: Sep  7, 2026 20:33:36.982120901 UTC
    Epoch Arrival Time: 1788813216.982120901
    [Time shift for this packet: 0.000000000 seconds]
    [Time delta from previous captured frame: 350.893 microseconds]
    [Time since reference or first frame: 610.561 microseconds]
    Frame Number: 4
    Frame Length: 149 bytes (1192 bits)
    Capture Length: 149 bytes (1192 bits)
    [Frame is marked: False]
    [Frame is ignored: False]
    [Protocols in frame: eth:ethertype:ip:tcp:http]
    Character encoding: ASCII (0)
Ethernet II, Src: 00:00:00_00:00:00 (00:00:00:00:00:00), Dst: 00:00:00_00:00:00 (00:00:00:00:00:00)
    Destination: 00:00:00_00:00:00 (00:00:00:00:00:00)
        .... ..0. .... .... .... .... = LG bit: Globally unique address (factory default)
        .... ...0 .... .... .... .... = IG bit: Individual address (unicast)
    Source: 00:00:00_00:00:00 (00:00:00:00:00:00)
```

Figure 9: *MAC Address Inspection & Layer 2 Ethernet Dissection* — Verifying all-zero MAC address structure (00:00:00:00:00:00) characteristic of kernel loopback.

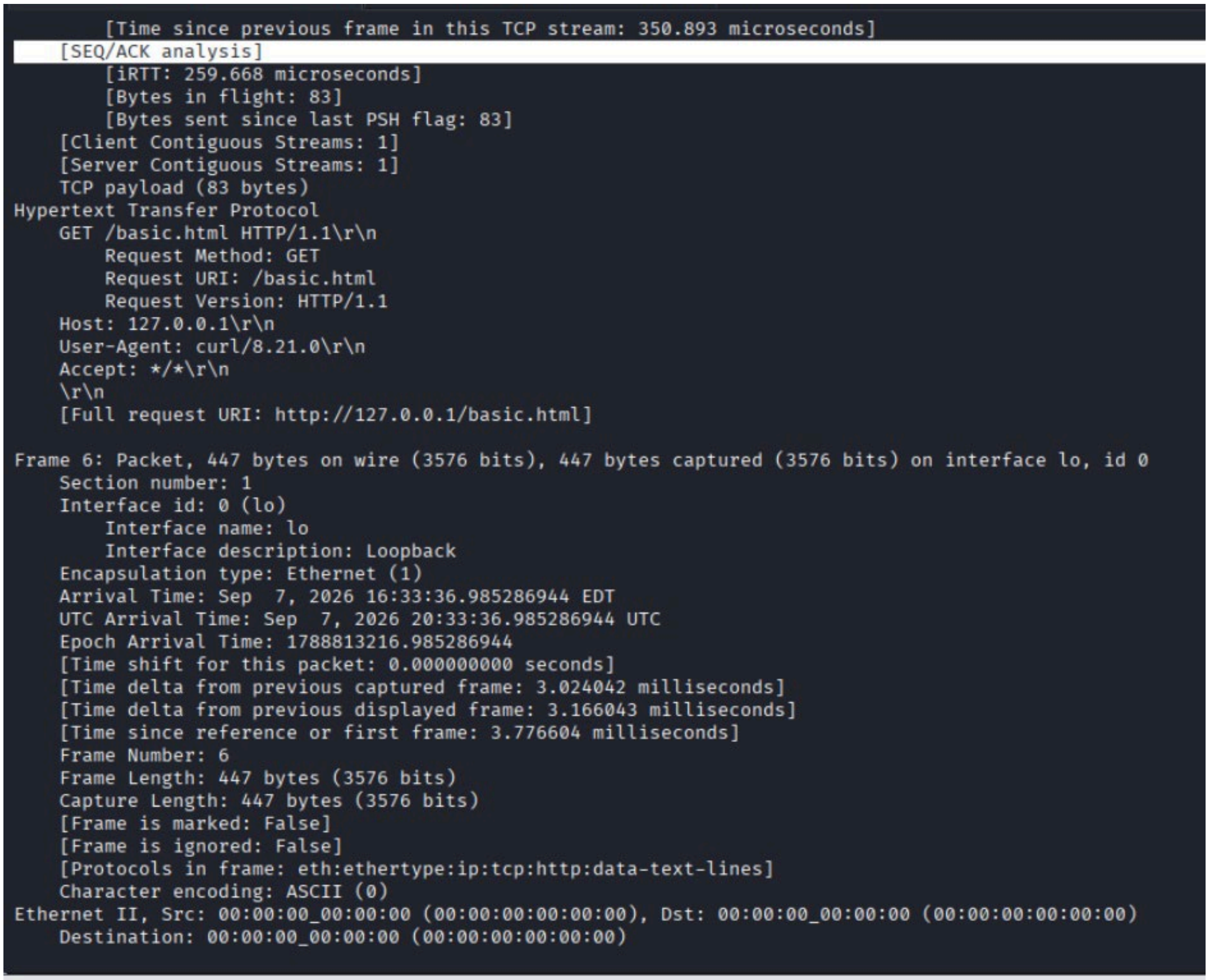


```
kali@kali: ~/SBT-DF203-Lab1    kali@kali: ~/SBT-DF203-Lab1
Ethernet II, Src: 00:00:00_00:00:00 (00:00:00:00:00:00), Dst: 00:00:00_00:00:00 (00:00:00:00:00:00)
    Destination: 00:00:00_00:00:00 (00:00:00:00:00:00)
        .... ..0. .... .... .... .... = LG bit: Globally unique address (factory default)
        .... ...0 .... .... .... .... = IG bit: Individual address (unicast)
    Source: 00:00:00_00:00:00 (00:00:00:00:00:00)
        .... ..0. .... .... .... .... = LG bit: Globally unique address (factory default)
        .... ...0 .... .... .... .... = IG bit: Individual address (unicast)
    Type: IPv4 (0x0800)
    [Stream index: 0]
Internet Protocol Version 4, Src: 127.0.0.1, Dst: 127.0.0.1
    0100 .... = Version: 4
    .... 0101 = Header Length: 20 bytes (5)
    Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
        0000 00.. = Differentiated Services Codepoint: Default (0)
        .... ..00 = Explicit Congestion Notification: Not ECN-Capable Transport (0)
    Total Length: 135
    Identification: 0x6923 (26915)
    010. .... = Flags: 0x2, Don't fragment
        0... .... = Reserved bit: Not set
        .1.. .... = Don't fragment: Set
        ..0. .... = More fragments: Not set
    ...0 0000 0000 0000 = Fragment Offset: 0
    Time to Live: 64
    Protocol: TCP (6)
    Header Checksum: 0xd34b [validation disabled]
    [Header checksum status: Unverified]
    Source Address: 127.0.0.1
    Destination Address: 127.0.0.1
    [Stream index: 0]
Transmission Control Protocol, Src Port: 34664, Dst Port: 80, Seq: 1, Ack: 1, Len: 83
    Source Port: 34664
    Destination Port: 80
    [Stream index: 0]
    [Stream Packet Number: 4]
    [Conversation completeness: Incomplete, ESTABLISHED (7)]
        ..0. .... = RST: Absent
        ...0 .... = FIN: Absent
        .... 0... = Data: Absent
        .... .1.. = ACK: Present
        .... ..1. = SYN-ACK: Present
        .... ...1 = SYN: Present
```

Figure 10: *Detailed Protocol Tree Inspection (Packet 4 - Header)* — Full packet dissection displaying IPv4 flags, TCP flags (SYN, ACK present), and 83-byte payload.

```
    .... ...1 = SYN: Present
    [Completeness Flags:█···ASS]
[TCP Segment Len: 83]
Sequence Number: 1    (relative sequence number)
Sequence Number (raw): 2501720227
[Next Sequence Number: 84    (relative sequence number)]
Acknowledgment Number: 1    (relative ack number)
Acknowledgment number (raw): 2758911253
1000 .... = Header Length: 32 bytes (8)
Flags: 0×018 (PSH, ACK)
    000. .... .... = Reserved: Not set
    ...0 .... .... = Accurate ECN: Not set
    .... 0... .... = Congestion Window Reduced: Not set
    .... .0.. .... = ECN-Echo: Not set
    .... ..0. .... = Urgent: Not set
    .... ...1 .... = Acknowledgment: Set
    .... .... 1... = Push: Set
    .... .... .0.. = Reset: Not set
    .... .... ..0. = Syn: Not set
    .... .... ...0 = Fin: Not set
    [TCP Flags: ·······AP···]
Window: 512
[Calculated window size: 65536]
[Window size scaling factor: 128]
Checksum: 0×fe7b [unverified]
[Checksum Status: Unverified]
Urgent Pointer: 0
Options: (12 bytes), No-Operation (NOP), No-Operation (NOP), Timestamps
    TCP Option - No-Operation (NOP)
        Kind: No-Operation (1)
    TCP Option - No-Operation (NOP)
        Kind: No-Operation (1)
    TCP Option - Timestamps: TSval 2818461323, TSecr 2818461323
        Kind: Time Stamp Option (8)
        Length: 10
        Timestamp value: 2818461323
        Timestamp echo reply: 2818461323
[Timestamps]
    [Time since first frame in this TCP stream: 610.561 microseconds]
    [Time since previous frame in this TCP stream: 350.893 microseconds]
[SEQ/ACK analysis]
```

Figure 11: *TCP Header Options & Flags Analysis (Packet 4)* — Detailed inspection of TCP segment flags (PSH, ACK - 0x018), 32-byte header, window scaling, and timestamp options.



```
        [Time since previous frame in this TCP stream: 350.893 microseconds]
    [SEQ/ACK analysis]
        [iRTT: 259.668 microseconds]
        [Bytes in flight: 83]
        [Bytes sent since last PSH flag: 83]
    [Client Contiguous Streams: 1]
    [Server Contiguous Streams: 1]
    TCP payload (83 bytes)
Hypertext Transfer Protocol
    GET /basic.html HTTP/1.1\r\n
        Request Method: GET
        Request URI: /basic.html
        Request Version: HTTP/1.1
    Host: 127.0.0.1\r\n
    User-Agent: curl/8.21.0\r\n
    Accept: */*\r\n
    \r\n
    [Full request URI: http://127.0.0.1/basic.html]

Frame 6: Packet, 447 bytes on wire (3576 bits), 447 bytes captured (3576 bits) on interface lo, id 0
    Section number: 1
    Interface id: 0 (lo)
        Interface name: lo
        Interface description: Loopback
    Encapsulation type: Ethernet (1)
    Arrival Time: Sep  7, 2026 16:33:36.985286944 EDT
    UTC Arrival Time: Sep  7, 2026 20:33:36.985286944 UTC
    Epoch Arrival Time: 1788813216.985286944
    [Time shift for this packet: 0.000000000 seconds]
    [Time delta from previous captured frame: 3.024042 milliseconds]
    [Time delta from previous displayed frame: 3.166043 milliseconds]
    [Time since reference or first frame: 3.776604 milliseconds]
    Frame Number: 6
    Frame Length: 447 bytes (3576 bits)
    Capture Length: 447 bytes (3576 bits)
    [Frame is marked: False]
    [Frame is ignored: False]
    [Protocols in frame: eth:ethertype:ip:tcp:http:data-text-lines]
    Character encoding: ASCII (0)
Ethernet II, Src: 00:00:00_00:00:00 (00:00:00:00:00:00), Dst: 00:00:00_00:00:00 (00:00:00:00:00:00)
    Destination: 00:00:00_00:00:00 (00:00:00:00:00:00)
```

Figure 12: *HTTP Request Details & Frame 6 Ingestion (Packets 4 & 6)* — Dissecting HTTP GET headers and transition to incoming HTTP/1.1 200 OK frame (447 bytes on wire).

```
Ethernet II, Src: 00:00:00_00:00:00 (00:00:00:00:00:00), Dst: 00:00:00_00:00:00 (00:00:00:00:00:00)
    Destination: 00:00:00_00:00:00 (00:00:00:00:00:00)
        .... ..0. .... .... .... .... = LG bit: Globally unique address (factory default)
        .... ...0 .... .... .... .... = IG bit: Individual address (unicast)
    Source: 00:00:00_00:00:00 (00:00:00:00:00:00)
        .... ..0. .... .... .... .... = LG bit: Globally unique address (factory default)
        .... ...0 .... .... .... .... = IG bit: Individual address (unicast)
    Type: IPv4 (0x0800)
    [Stream index: 0]
Internet Protocol Version 4, Src: 127.0.0.1, Dst: 127.0.0.1
    0100 .... = Version: 4
    .... 0101 = Header Length: 20 bytes (5)
    Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
        0000 00.. = Differentiated Services Codepoint: Default (0)
        .... ..00 = Explicit Congestion Notification: Not ECN-Capable Transport (0)
    Total Length: 433
    Identification: 0xf06f (61551)
    010. .... = Flags: 0x2, Don't fragment
        0... .... = Reserved bit: Not set
        .1.. .... = Don't fragment: Set
        ..0. .... = More fragments: Not set
    ...0 0000 0000 0000 = Fragment Offset: 0
    Time to Live: 64
    Protocol: TCP (6)
    Header Checksum: 0x4ad5 [validation disabled]
    [Header checksum status: Unverified]
    Source Address: 127.0.0.1
    Destination Address: 127.0.0.1
    [Stream index: 0]
Transmission Control Protocol, Src Port: 80, Dst Port: 34664, Seq: 1, Ack: 84, Len: 381
    Source Port: 80
    Destination Port: 34664
    [Stream index: 0]
    [Stream Packet Number: 6]
    [Conversation completeness: Incomplete, DATA (15)]
        ..0. .... = RST: Absent
        ...0 .... = FIN: Absent
        .... 1... = Data: Present
        .... .1.. = ACK: Present
        .... ..1. = SYN-ACK: Present
        .... ...1 = SYN: Present
```

Figure 13: *Ethernet, IPv4 & TCP Header Dissection (Packet 6)* — Detailed inspection of server response packet showing total length 433 bytes and TCP segment length 381 bytes.

```
        .... ..1. = SYN-ACK: Present
        .... ...1 = SYN: Present
        [Completeness Flags: ··DASS]
    [TCP Segment Len: 381]
    Sequence Number: 1    (relative sequence number)
    Sequence Number (raw): 2758911253
    [Next Sequence Number: 382    (relative sequence number)]
    Acknowledgment Number: 84    (relative ack number)
    Acknowledgment number (raw): 2501720310
    1000 .... = Header Length: 32 bytes (8)
    Flags: 0×018 (PSH, ACK)
        000. .... .... = Reserved: Not set
        ...0 .... .... = Accurate ECN: Not set
        .... 0... .... = Congestion Window Reduced: Not set
        .... .0.. .... = ECN-Echo: Not set
        .... ..0. .... = Urgent: Not set
        .... ...1 .... = Acknowledgment: Set
        .... .... 1... = Push: Set
        .... .... .0.. = Reset: Not set
        .... .... ..0. = Syn: Not set
        .... .... ...0 = Fin: Not set
        [TCP Flags: ·······AP···]
    Window: 512
    [Calculated window size: 65536]
    [Window size scaling factor: 128]
    Checksum: 0×ffa5 [unverified]
    [Checksum Status: Unverified]
    Urgent Pointer: 0
    Options: (12 bytes), No-Operation (NOP), No-Operation (NOP), Timestamps
        TCP Option - No-Operation (NOP)
            Kind: No-Operation (1)
        TCP Option - No-Operation (NOP)
            Kind: No-Operation (1)
        TCP Option - Timestamps: TSval 2818461326, TSecr 2818461323
            Kind: Time Stamp Option (8)
            Length: 10
            Timestamp value: 2818461326
            Timestamp echo reply: 2818461323
    [Timestamps]
        [Time since first frame in this TCP stream: 3.776604 milliseconds]
        [Time since previous frame in this TCP stream: 3.024042 milliseconds]
```

Figure 14: *TCP Response Flags & Timestamp Analysis (Packet 6)* — Analysis of server response flags (PSH, ACK - 0x018), relative sequence/acknowledgment numbers, and TCP timestamp echo.

```
            Length: 10
            Timestamp value: 2818461326
            Timestamp echo reply: 2818461323
    [Timestamps]
        [Time since first frame in this TCP stream: 3.776604 milliseconds]
        [Time since previous frame in this TCP stream: 3.024042 milliseconds]
    [SEQ/ACK analysis]
        [iRTT: 259.668 microseconds]
        [Bytes in flight: 381]
        [Bytes sent since last PSH flag: 381]
    [Client Contiguous Streams: 1]
    [Server Contiguous Streams: 1]
    TCP payload (381 bytes)
Hypertext Transfer Protocol
    HTTP/1.1 200 OK\r\n
        Response Version: HTTP/1.1
        Status Code: 200
        [Status Code Description: OK]
        Response Phrase: OK
    Date: Mon, 07 Sep 2026 20:33:36 GMT\r\n
    Server: Apache/2.4.68 (Debian)\r\n
    Last-Modified: Mon, 07 Sep 2026 20:19:35 GMT\r\n
    ETag: "82-65aea5637199b"\r\n
    Accept-Ranges: bytes\r\n
    Content-Length: 130\r\n
        [Content length: 130]
    Vary: Accept-Encoding\r\n
    Content-Type: text/html\r\n
    \r\n
    [Request in frame: 4]
    [Time since request: 3.166043 milliseconds]
    [Request URI: /basic.html]
    [Full request URI: http://127.0.0.1/basic.html]
    File Data: 130 bytes
Line-based text data: text/html (2 lines)
    <!DOCTYPE html>\n
    <html><body><h1>SBT-DF203 HTTP Evidence</h1><p>Name: Almustapha Yusuf</p><p>Reg No: fwsd2511343</p></body></html>\n

┌──(kali㉿kali)-[~/SBT-DF203-Lab1]
└─$ 
```

Figure 15: *HTTP 200 OK Response Header & HTML Payload Dissection (Packet 6)* — Inspection of HTTP status code 200, Content-Type text/html, Content-Length 130 bytes, and reconstructed HTML evidence payload.

8. Findings, Limitations, and Conclusion

- **Forensic Findings:** The HTTP server functioned normally, returning a 200 OK status code. The full transaction was reconstructed in plaintext without encryption. The TCP session followed standard RFC 793 state transitions from SYN through FIN-ACK closure.
- **Forensic Limitations:** Loopback captures lack physical Layer 2 Ethernet telemetry. To examine realistic MAC address spoofing, ARP Poisoning, or router hop artifacts, a multi-host virtual network (e.g., Host-Only or Bridged adapter) must be utilized.
- **Conclusion:** This laboratory successfully demonstrated foundational network forensics principles: packet capture using CLI engines (TShark), forensic integrity preservation through SHA-256 hashing, TCP session reconstruction, and deep OSI encapsulation analysis.

9. Appendix of Commands and Hashes

```
# Packet Capture on Loopback
sudo tshark -i lo -f 'tcp port 80' -w evidence/basic.pcapng

# Controlled HTTP Client Generation
curl --no-keepalive -v http://127.0.0.1/basic.html

# Integrity Hashing
sha256sum evidence/basic.pcapng working/basic_working.pcapng

# Dissect Handshake & HTTP Fields
tshark -r working/basic_working.pcapng -Y 'tcp.flags.syn==1' -T fields ...
tshark -r working/basic_working.pcapng -Y 'http.request' -T fields ...
tshark -r working/basic_working.pcapng -q -z follow,tcp,ascii,0
```

Evidence Integrity: Cryptographic Hash Validation:

- **Source Evidence Hash:** 4fb3910cb6127763a6367512e74688e3ed779c345b4f0da1e83f51d4a79d6e3 (evidence/basic.pcapng)
- **Working Duplicate Hash:** 4fb3910cb6127763a6367512e74688e3ed779c345b4f0da1e83f51d4a79d6e3 (working/basic_working.pcapng)

Student Declaration

```
I confirm that I completed this lab in an authorized environment, preserved the evidence, did not
target any third-party system, and accurately documented my own commands, observations, and
conclusions.

Student Signature: Almustapha Yusuf                      Date: 07/09/2026
```